Po-Sheng Cheng

Master of Industrial Design, Rhode Island School of Design, exp. Jun. 2026
Cross-enrolled at School of Engineering, Brown University
B.Sc. Bio-Mechatronics Engineering, National Taiwan University, Jan 2023

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Portfolio: https://bencer3283.github.io/art/

Skills

- Electrical: PCB design (Altium), STM32, ESP32, Raspberry Pi, system interface (I2C, SPI, UART)
- Coding: C/C++, JavaScript (React, Express, MongoDB), Dart (Flutter), Python (PyTorch, NumPy), Git, Linux

Experiences

Product Development Apprentice, Google, Jun. - Sep. 2024 — Taipei, Taiwan

- Led a team of 5 to design a Hardware-as-a-Service platform for managing lost items in public facilities. Laid out the system architecture of the hardware kiosk, user-facing app and web services.
- My prototypes of Raspberry Pi and ESP32 hardware, a React-based website, a mobile app and a RESTful API service drove the development direction from a single hardware product to an integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022 — Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating an Arduino microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.
- The prototype recieved widespread praise across many departments (pm, engineering, design) in the company while I worked with them to support guiding the direction of product development.

College Student Researcher, NTU, Jul. 2021 - Feb. 2022 — Taipei, Taiwan

- Developed a novel, integrated spectral mapping system that vastly accelerates scientific spectral measurement.
- By using linear dispersion optics and optimizing the EMCCD readout algorithm in LabVIEW and C++, the scanning time was reduced by 26%.
- Awarded the **2021 Technology Innovation Award** by CCMS, NTU and **College Student Research Creativity Award** by National Science and Technology Council of Taiwan with this project.

Projects

Haptic Belt

- Designed a custom rigid-flex PCB for a belt that uses vibration feedback to provide navigation for visually impaired people.
- Performed tesing on driver and interfacing ICs and vibration motors using oscilloscope. Implemented STM32 firmware for the vibration motor driver.

MITM Router

- Built a WIFI router that brings physical, tactile interactions to the user's internet usage with Linux networking stack on a Raspberry Pi.
- Extensive prototyping and user testing of the physical user interface was done using several sensors/actuators on a RP2040 microcontroller.

PageMate smart bookmark

- Built the physical device with advanced touch gesture sensing on ESP32 and the Node.js (Express) web service of a smart bookmark that can display reading progress of the user's ebooks.
- Used Figma to create a demo experience that can be updated in real time using my Node.js web service.